

Recall from the last few classes:

- We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by

$$\det(A) = ad - bc.$$

- **Theorem:** If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.
- **Theorem:** Suppose A and B are n by n invertible matrices, then the following hold.
 1. A^{-1} is invertible and $(A^{-1})^{-1} = A$.
 2. $(AB)^{-1} = B^{-1}A^{-1}$.
 3. $(A^T)^{-1} = (A^{-1})^T$.
- An **elementary matrix** E is a matrix obtained from the identity matrix by a single elementary row operation.
- **Theorem:** A is invertible if and only if A is row equivalent to the identity matrix. Furthermore, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

1.

(a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. Find A^{-1} if possible.

(b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. Find A^{-1} if possible.

Invertible Matrix Theorem Let A be an $n \times n$ matrix, then the following are all equivalent.

- (a) A is invertible.
- (b) A is row-equivalent to I_n .
- (c) A has n pivot positions.
- (d) $A\vec{x} = \vec{0}$ has only the trivial solution.
- (e) The columns of A are linearly independent.
- (f) $x \mapsto A\vec{x}$ is one-to-one (injective).
- (g) For all $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ is consistent.
- (h) The columns of A span $\mathbb{R}^n$.
- (i) $\vec{x} \mapsto A\vec{x}$ is onto (surjective).
- (j) There exists a matrix C such that $CA = I_n$.
- (k) There exists a matrix D such that $AD = I_n$.
- (l) A^T is invertible.

2. Prove the theorem.

3. Determine if the following matrices are invertible.

(a) Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & -4 & 0 \\ 8 & 5 & -3 \end{bmatrix}$.

(b) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Theorem: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear and let A be the associated matrix. Then T is invertible if and only if A is invertible.

4. Prove the theorem.

Goal: Develop a numerical test for invertibility.

5. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Find a condition on the entries of A for A to be invertible.

Definition 1. For a matrix A define A_{ij} to be the matrix obtained from A by deleting row i and column j .

Remark:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \det A_{11} - b \det A_{12} + c \det A_{13}.$$

Definition 2. For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).$$

6. Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Definition 3. The (i, j) -**cofactor** of A is given by $c_{ij} = (-1)^{i+j} \det A_{ij}$, and the **cofactor expansion along row i** is given by $\sum_{j=1}^n a_{ij} c_{ij}$.

Theorem 3.1: Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.

7. Compute the determinant of $C = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}$.

Definition 4. Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is **upper triangular** if $a_{ij} = 0$ when $i > j$.

Theorem 3.2: Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11}a_{22} \dots a_{nn} = \prod_{j=1}^n a_{jj}.$$

8. Prove the theorem.